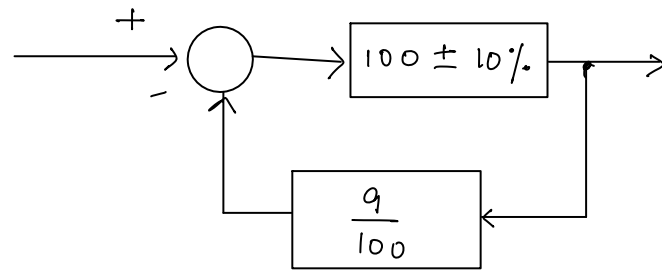


Tutorial 12

Q1.) A negative feedback system as shown in fig 1 has an amplifier of gain 100 with $\pm 10\%$ tolerance in the forward path, and an attenuator of value $9/100$ in the feedback path. The overall system gain is approximately:

Gain of a negative feedback system is:

$$G = \frac{A}{1 + A\beta}$$



A = open loop gain of the amplifier = $100 \pm 10\%$.

β = feedback factor = $\frac{9}{100}$

$$A_{\min} = 100 - 10\% = 100 - \frac{10}{100} \times 100 = 90$$

$$A_{\max} = 100 + 10\% = 100 + \frac{10}{100} \times 100 = 110$$

$$\beta = \frac{9}{100} = 0.09$$

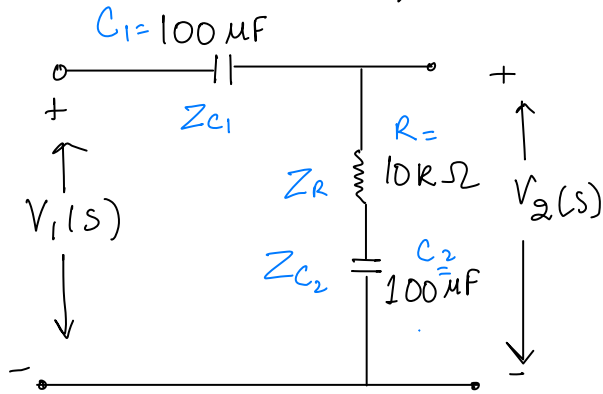
$$G_1 = \frac{A_{\min}}{1 + A_{\min}\beta} = \frac{90}{1 + 90 \times 0.09} \approx 9.89 \approx 9.9$$

$$G_2 = \frac{A_{\max}}{1 + A_{\max}\beta} = \frac{110}{1 + 110 \times 0.09} \approx 10.09 \approx 10.1$$

Overall system gain is around 10

$$\approx 10 \pm 1\%$$

Q.2. The transfer function $\frac{V_2(s)}{V_1(s)}$ of the circuit shown below is:-



$$Z_{C1} = \frac{1}{sC_1} \quad ; \quad Z_{C2} = \frac{1}{sC_2}$$

$$Z_R = R$$

Use V/g division :-

$$V_2(s) = \frac{(Z_R + Z_{C2})}{Z_{C1} + (Z_R + Z_{C2})} V_1(s)$$

$$Z_{C1} = \frac{1}{sC_1} = \frac{1}{s \times 100 \times 10^{-6}} = \frac{1}{0.0001s}$$

$$Z_{C2} = \frac{1}{sC_2} = \frac{1}{s \times 100 \times 10^{-6}} = \frac{1}{0.0001s}$$

$$Z_R = R = 10 \times 10^3$$

$$\therefore Z_R + Z_{C2} = 10^4 + \frac{1}{0.0001s}$$

$$\therefore \frac{V_2(s)}{V_1(s)} = \frac{10^4 + \frac{1}{0.0001s}}{\frac{1}{0.0001s} + 10^4 + \frac{1}{0.0001s}}$$

$$= \frac{1 + 10^4(0.0001s)}{2 + 10^4(0.0001s)}$$

$$T(s) = \frac{s + 1}{s + 2}$$

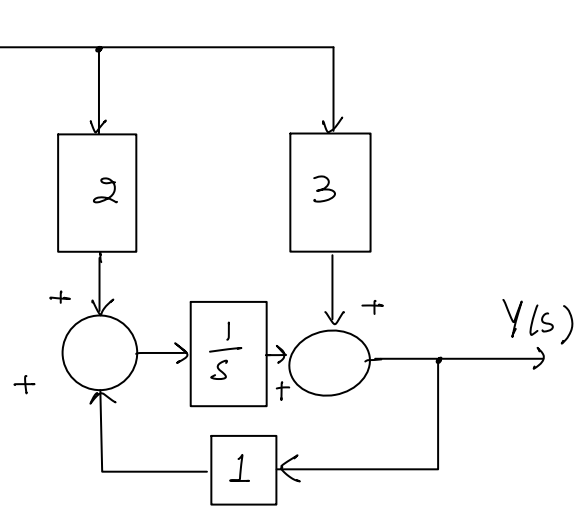
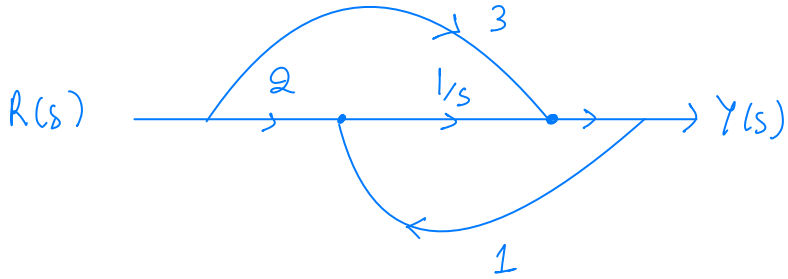
Q3. For the block diagram shown, the transfer function $\frac{Y(s)}{R(s)}$ is

```

graph LR
    R["R(s)"] --> Sum((+))
    Sum --> B1["1/s"]
    B1 --> B2["1/s"]
    B2 --> Y["Y(s)"]
    Y --> B3["1/s"]
    B3 --> Sum
  
```

Reduce the block diag to find $\frac{Y(s)}{R(s)}$

Form a signal flow graph (sfg)



There are two forward paths:

$$P_1 = 3, \Delta_1 = 1$$

$$P_2 = 2 \times \frac{1}{5}, \quad \Delta_2 = 1$$

Loops, $L_1 = 1 \times 1 \times 1 = 1$

Mason's gain formula

$$\frac{Y(s)}{R(s)} = \frac{P_1 \Delta_1 + P_2 \Delta_2}{\Delta} = \frac{3 \times 1 + \frac{2}{s} \times 1}{1 - L_1} = \frac{3s+2}{s-1}$$

$$T(s) = \frac{3s + 2}{s - 1}$$

Mason's Rule: Transfer function T_i of a system represented by a signal flow graph is

$$T = \frac{C(s)}{D(s)} = \frac{\sum_{i=1}^m P_i \Delta_i}{\Delta}$$

$$T = \frac{C(S)}{R(S)} = \frac{\sum_{i=1}^m p_i \Delta_i}{\Delta}$$

where, n = no. of forward paths, P_i = i th forward-path gain

$$\Delta = 1 - (\text{Sum of loop gains of all individual loops}) + (\text{sum of gain products of all possible combinations of non-touching loops}) - (\text{" " " " "all 3...})$$

Δ_i : value of Δ for part of the graph not touching with the i th forward path.

Q4. Given the magnetic circuit shown below has dimensions
 $A_c = A_g = 9 \text{ cm}^2$, $g = 0.05 \text{ cm}$, $l_c = 30 \text{ cm}$,

$N = 500$ turns $\rightarrow \mu_r = 70,000$ for core material,
 find a) the reluctance R_c and R_g
 b) flux and
 c) the current i ($B_c = 1.0 \text{ T}$) { $i = 0.8 \text{ A}$ }

$$R_c = \frac{l_c}{\mu A_c} = \frac{l_c}{\mu_0 \mu_r A_c} \quad \text{and} \quad R_g = \frac{l_g}{\mu_0 A_g}$$

$$R_c = \frac{0.3}{(70,000 \times 4\pi \times 10^{-7}) \times (9 \times 10^{-4})} = 3.79 \times 10^3 \text{ AT/Wb}$$

$$R_g = \frac{g}{\mu_0 A_g} = \frac{5 \times 10^{-4}}{(4\pi \times 10^{-7}) \times (9 \times 10^{-4})} = 4.42 \times 10^5 \text{ AT/Wb}$$

$$\phi = B_c A_c = 1 [9 \times 10^{-4}] = 9 \times 10^{-4} \text{ Wb}$$

$$F = \phi (R_c + R_g) = \phi (R_{\text{total}}) \approx 401 \text{ AT}$$

$$F = Ni$$

$$i = \frac{F}{N} = \frac{(9 \times 10^{-4}) [4.46 \times 10^5]}{500} = 0.8 \text{ A}$$

Q5) For the iron core, $a = 5 \text{ cm}$, $b = 20 \text{ cm}$, $c = 10 \text{ cm}$, $d = 20 \text{ cm}$,
 $N = 200$. Find the current required to establish 6.5 mWb .

a) across 5 mm air gap

b) if there is no air gap (given $\mu_r = 4138$)

Area of cross-section of the core, $A_c = a \times c$

$$= (5 \times 10^{-2}) (10 \times 10^{-2}) = 5 \times 10^{-3} \text{ m}^2$$

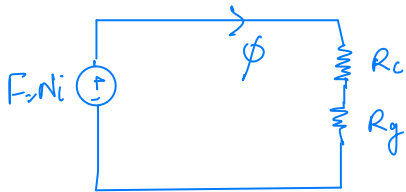
length of the magnetic path, $l_c = 2b + 2d - \text{air gap}$.

$$= 2(20 \times 10^{-2}) + 2(20 \times 10^{-2}) - l_g$$

$$= 80 \times 10^{-2} \text{ m} - \underbrace{0.005}$$

$$= 0.8 \text{ m}$$

Since small
compared to $80 \times 10^{-2} \text{ m}$
 $\therefore$ neglected



Reluctance of iron core

$$R_{\text{core}} = \frac{l_c}{\mu_r \mu_0 A_c} = \frac{0.8}{4138 \times 4\pi \times 10^{-7} \times 5 \times 10^{-3}} \approx 30718.5 \text{ AT/Wb}$$

$$R_g = \frac{l_g}{\mu_0 A_g} = \frac{5 \times 10^{-3}}{4\pi \times 10^{-7} \times 5 \times 10^{-3}} = 795774.7 \text{ AT/Wb}$$

$$R_{\text{total}} = R_{\text{core}} + R_g \approx 826493.2 \text{ AT/Wb}$$

$$I = \frac{\phi R_{\text{total}}}{N} = \frac{6.5 \times 10^{-3} \times R_{\text{total}}}{200} \approx 26.86 \text{ A}$$

b) Without air gap.

$$R_{\text{total}} = R_{\text{core}} = 30718.5 \text{ AT/Wb}$$

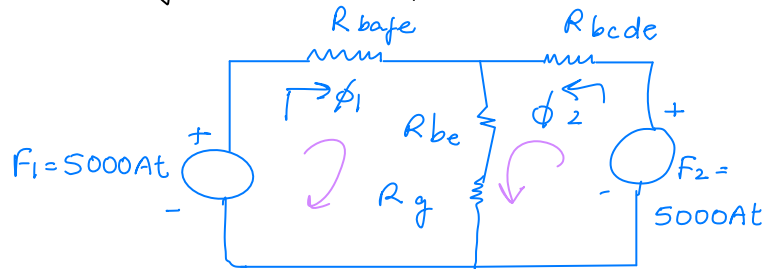
$$I = \frac{\phi R_{\text{total}}}{N} = \frac{6.5 \times 10^{-3} \times R_{\text{total}}}{200} \approx 1 \text{ A}$$

Q6. In the magnetic circuit shown in the figure, the relative permeability of the ferromagnetic material is 1200. Neglect magnetic leakage and fringing. All dimensions are in cm, and the magnetic material has a square CSA. Determine

- air gap flux
- air gap flux density
- Magnetic field intensity in the air gap

Sol:-

$$\begin{aligned} F_1 &= N_1 I_1 \\ &= 500 \times 10 \\ &= 5000 \text{ At} \end{aligned}$$



$$\begin{aligned} F_2 &= N_2 I_2 \\ &= 500 \times 10 \\ &= 5000 \text{ At} \end{aligned}$$

$$\mu_c = \mu_r \mu_0 = (1200 \times 4\pi \times 10^{-7})$$

$$\begin{aligned} R_{bafc} &= \frac{l_{bafc}}{\mu_c A_c} = \frac{(52+52+52) \times 10^{-2}}{\mu_c \times 4 \times 10^{-4}} \\ &= 2.58 \times 10^6 \text{ At/Wb} \end{aligned}$$

Since depth = 2
Area of cross-section
= 2×2
= 4

$$\begin{aligned} R_{bcde} &= \frac{l_{bcde}}{\mu_c A_c} = \frac{(52+52+52) \times 10^{-2}}{\mu_c \times 4 \times 10^{-4}} \\ &= 2.58 \times 10^6 \text{ At/Wb} \end{aligned}$$

$$\begin{aligned} R_{be} &= \frac{l_{be}}{\mu_c A_c} = \frac{(52-0.5) \times 10^{-2}}{\mu_c \times 4 \times 10^{-4}} \\ &= 0.82 \times 10^6 \text{ At/Wb} \end{aligned}$$

$$R_g = \frac{l_g}{\mu_0 A_g} = \frac{0.5 \times 10^{-2}}{4\pi \times 10^{-7} \times 4 \times 10^{-4}} = 9.94 \times 10^6 \text{ At/Wb}$$

loop equations:-

loop 1:

$$-N_1 i_1 + \phi_1 R_{bafc} + \{ \phi_1 + \phi_2 \} \{ R_{be} + R_g \} = 0$$

$$\phi_1 \{ R_{bafc} + R_{be} + R_g \} + \phi_2 \{ R_{be} + R_g \} = N_1 i_1 \quad \text{--- (I)}$$

loop 2:

$$-N_2 i_2 + \phi_2 R_{bcde} + (\phi_2 + \phi_1) \{ R_{be} + R_g \} = 0$$

$$\phi_1 \{ R_{be} + R_g \} + \phi_2 \{ R_{be} + R_g + R_{bcde} \} = N_2 i_2 \quad \text{--- (II)}$$

After substituting R_{bcde} , R_{be} , R_g , R_{bafc} we get (III) & (IV),

$$13.34 \times 10^6 \phi_1 + 10.76 \times 10^6 \phi_2 = 5000 \quad \text{--- (III)}$$

$$10.76 \times 10^6 \phi_1 + 13.34 \times 10^6 \phi_2 = 5000 \quad \text{--- (IV)}$$

On solving, we get

$$\phi_1 = 207.3 \times 10^{-6} \text{ Wb}$$

$$\phi_2 = 207.3 \times 10^{-6} \text{ Wb}$$

i) Air gap flux $\phi_g = \phi_1 + \phi_2 = 414.6 \times 10^{-6} = 4.146 \times 10^{-4} \text{ Wb}$

ii) Air gap flux density

$$B_g = \frac{\phi_g}{A_g} = 1.0365 \text{ T} \quad (\text{or } \text{Wb/m}^2)$$

iii) Magnetic field intensity in the air gap is

$$H_g = \frac{B_g}{\mu_0} = 0.824 \times 10^6 \text{ At/m.}$$

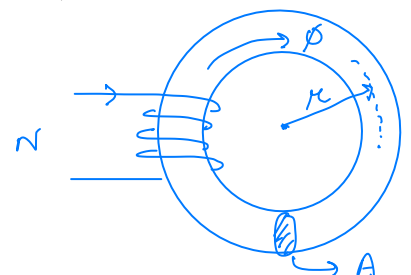
Q7.) It is desired to establish a flux of 0.5 mWb in a toroidal iron core with 400 turns and relative permeability $= 1000$.

The iron core has cross-sectional area of 10^{-3} m^2 and $R = 0.04 \text{ m}$.
Find the reluctance and the current required.

Sol:- Reluctance of a magnetic circuit

$$R = \frac{l}{\mu A}$$

$$l = 2\pi r = 2\pi(0.04) = 0.08\pi$$



$$\mu = \mu_r \mu_0 = 1000 \times 4\pi \times 10^{-7} = 4\pi \times 10^{-4}$$

$$A = 10^{-3} \text{ m}^2$$

$$R = \frac{0.08 \pi}{4\pi \times 10^{-4} \times 10^{-3}} = 2 \times 10^5$$

$$F = \phi R, \quad N i = \phi R, \\ \Rightarrow i = \frac{\phi R}{N} = \frac{0.5 \times 10^{-3} \times 2 \times 10^5}{4 \times 10^2} = 0.25 \text{ A}$$

- Q8.) A 10KVA 1 ϕ Xmer has a Xmatin ratio = 0.083 & is supplied from a 2.4kV supply. Neglecting losses, determine:
- full load I_2
 - min value of load R_L which can be connected across the sec. winding w/o the kVA rating being exceeded.
 - I_1

Sol:- $S = 10 \text{ KVA} = 10,000 \text{ VA}$

$$a = 0.083$$

$$V_1 = 2.4 \text{ kV} = 2400 \text{ V}$$

$$a = \frac{V_2}{V_1} = \frac{2400}{V_1} \Rightarrow V_1 = 199.2 \text{ V}$$

$$I_2 = \frac{S}{V_2} = \frac{10000}{199.2} = 50.2 \text{ A}$$

$$R_{\min} = \frac{V_2}{I_2} = \frac{199.2}{50.2} = 3.97 \Omega$$

$$I_1 = a I_2 = 4.17 \text{ A}$$

- Q9.) An ideal Xmer is used inside a DELL laptop charger (20V, 65W Output). Design this xmer: determine the winding ratio and the Ampere rating of the primary side cable connecting the charger to the wall socket supply at your home.

Output : $V_s = 20$
 $P_s = 65W$

Input : Assuming standard household voltage $V_p = 230V$

For ideal xmer, $P_p = P_s$ {no loss}

$$n = \frac{V_p}{V_s} = \frac{N_p}{N_s}$$

$$n = \frac{230}{20} = 11.5$$

$$P_s = V_s I_s$$

$$I_s = \frac{65}{20}$$

$$11.5 = \frac{I_s}{I_p} \quad , \quad I_p = \frac{I_s}{11.5} = 0.2826 \text{ A}$$